6. Memory (RAM) :-

> Checker strategy :

→ Reference model :- logic [$\overset{\text{data\_width}}{\widehat{DW}}$-1:0] ram_model [$\overset{\text{addr\_width}}{\widehat{AW}}$-1:0]

write & read are monitored.

→ Use assertions on interface to check for read & write enable & setup & hold times.

> Directed tests :-

→ Write & read random patterns at random addresses.

→ Check at boundary addresses (0 & $2^{\wedge AW}-1$).

→ Continuous write.

→ Continuous read.

→ Read & write at same address.

→ Read & write at same address simultaneously.

> Coverage :-

Bins for
i) min_addr = 0
ii) max_addr = $2^{\wedge AW}-1$
iii) mid_addr = $1 : 2^{\wedge AW}-2$
iv) Command_transitions = W→W, R→R, W→R, R→W.
v) Data
vi) Cross addr X transitions